

MathScript: Benchmarking MLLMs on Non-linear Layouts and Hallucinations in Authentic Handwritten Mathematics

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Abstract

The automatic structured extraction of handwritten responses on paper-based mathematics exams is fundamental to intelligent education, yet it remains highly challenging in real-world scenarios due to complex non-linear spatial layouts and severe visual noise. Existing benchmarks primarily focus on relatively clean documents or generic layout analysis, overlooking both the end-to-end reading order of complex handwritten mathematical content and the ability of models to actively refuse incomplete or degraded inputs. We introduce MathScript, the first fine-grained benchmark for information extraction from authentic high school mathematics exam handwritten responses. MathScript contains 446 carefully curated questions featuring real-world physical artifacts (such as multi-column interleaving and local insertions) and explicitly includes unrecognizable samples to evaluate active refusal behavior. We propose a multi-dimensional evaluation framework that comprehensively covers text extraction accuracy, formula recognition, sequence-level reading order, and intelligent refusal capability. Experiments on state-of-the-art multimodal large language models (MLLMs) demonstrate that although end-to-end models achieve strong content extraction performance, they universally suffer from structural parsing failures in complex non-linear layouts and consistently fail to refuse illegible inputs, instead generating confident but visually unfounded hallucinatory outputs. To address these challenges, we further propose a semantics-guided Reading order Repair Agent (RRA) and a reliability-aware mechanism, significantly improving structural consistency and reducing hallucination risks. These results highlight a critical shortcoming in current MLLMs and establish MathScript as a core benchmark for assessing model reliability under complex handwritten conditions. Our code and data are available in a GitHub repository.

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1 Introduction

In intelligent education scenarios, the automatic digitization of paper-based examinations and assignments serves as a crucial foundation for intelligent grading, question bank construction, and

learning analytics. To effectively support these applications, Optical Character Recognition (OCR) is evolving from a task centered on text transcription into a broader document understanding problem. The rapid development of Multimodal Large Language Models (MLLMs) has further advanced this transformation by integrating visual perception, layout understanding, and linguistic reasoning within a unified framework. This development offers new opportunities for handling complex educational documents, but demands stronger structural and spatial understanding.

While existing MLLMs have shown impressive semantic parsing capabilities on relatively regular documents, their deployment in real-world educational scenarios remains challenging. In real-world settings, MLLMs are required to process handwritten documents with highly complex layouts and irregular spatial organization. Compared to standard printed text, authentic handwritten response pages exhibit extreme characteristics such as unstable handwriting, blurred region boundaries, interleaved distribution of multi-column content, and non-linear reading orders. Furthermore, the dense interleaving of text and mathematical formulas exacerbates the complexity of structural parsing. Collectively, these factors suggest that OCR for complex handwritten layouts requires MLLMs not only to accurately "recognize what is seen," but also to possess robust structural modeling capabilities to determine "what to read first and what to read next". Moreover, MLLMs must be capable of making conservative judgments—namely, "whether to refuse to output"—when visual evidence is insufficient.

Despite the abundance of OCR evaluation efforts within the community, the coverage of existing benchmarks for complex handwritten layout scenarios remains highly limited. First, most current benchmarks use locally cropped regions, text blocks, or entire pages as their basic units, focusing heavily on final recognition accuracy while lacking a fine-grained assessment of the end-to-end reading order. Consequently, they struggle to accurately reflect sequence errors unique to complex non-linear layouts, such as content crosstalk and reading chain disruption caused by region interleaving or cross-region insertions. Second, existing evaluations fail to explicitly distinguish between "content recognition errors" and "sequence recognition errors," compressing them into a single error category, which obscures critical reading order issues within the overall score. More importantly, previous studies primarily focus on text accuracy, lacking systematic analysis of hallucinated outputs and refusal behaviors produced by models under low-quality inputs (e.g., blur, occlusion). This prevents a comprehensive assessment of model reliability in real-world applications.

Motivated by these observations, we propose MathScript, a fine-grained evaluation benchmark specifically designed to assess OCR performance on complex handwritten layouts. Compared to existing works, MathScript focuses on real-world educational scenarios, systematically covering key challenges such as non-linear layouts, region interleaving, cross-block dependencies, and the mixture of handwritten text and mathematical formulas. Within a unified

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evaluation framework, we evaluate not only the final content extraction performance but also explicitly introduce sequence-level reading order metrics and hallucination/refusal-related metrics, thereby deeply characterizing the MLLMs' structural understanding capabilities and output reliability. Systematic experiments on MathScript reveal two core failure bottlenecks in current MLLMs: First, structural parsing failure—when the layout transitions from linear to non-linear organization, models are highly prone to severe sequence errors at local interleavings and cross-region jumps. Second, a lack of output reliability—when faced with insufficient visual evidence, models frequently generate uncontrolled structural hallucinations and unfounded text completions.

To verify the practical guiding significance of this benchmark for subsequent method design, we further propose two benchmark-driven improvement strategies. For structural parsing in complex layouts, we design a semantics-guided Reading order Repair Agent (RRA), which utilizes the semantic consistency judgments of large models to correct candidate reading orders within a restricted action space. For the reliability challenge posed by low-evidence inputs, we design a reliability-aware OCR pipeline, introducing an upfront legibility-gated routing mechanism to filter high-risk inputs and mitigate hallucination risks. Experimental results demonstrate that these strategies achieve stable performance gains on the benchmark. The main contributions of this paper are as follows:

- We propose **MathScript**, the first fine-grained benchmark for evaluating MLLMs on the OCR of complex handwritten layouts in high school mathematics. Focusing on complex non-linear layouts in real educational scenarios, this benchmark employs sequence-level fine-grained analysis, revealing that reading order errors in math exams are primarily caused by text interleaving. Furthermore, it decouples OCR accuracy from reading order evaluation, significantly enhancing the validity and granularity of the assessment.
- We integrate reading order and output reliability into a unified, multi-dimensional evaluation framework to systematically assess the capabilities of mainstream MLLMs in handwritten OCR for high school math exams. Concurrently, we identify two core failure modes of current MLLMs in this task: parsing errors under complex structures, and the imbalance between hallucination and refusal when visual evidence is insufficient.
- We propose two improvement strategies, including a semantics-guided Reading order Repair Agent (RRA) and an OCR process decoupling mechanism, which effectively enhance the reading order recognition capabilities of MLLMs in complex layout scenarios and significantly reduce hallucinations. This effectively validates that MathScript can not only accurately diagnose model bottlenecks but also practically guide the optimization and performance enhancement of subsequent methods.

2 Related Works

2.1 Reading Order Measurement

Reading-order recovery is a core problem in document understanding [13]. It aims to reconstruct content sequences that follow human reading habits after detecting text lines, blocks, or layout regions. Early approaches rely on geometric heuristics, such as

top-to-bottom or left-to-right rules. These methods are simple, interpretable, and effective on regular layouts. Later work formulates reading-order recovery as a learnable sorting or relational prediction task [18]. With the development of document understanding models such as LayoutLMv2 [19], LayoutLMv3 [5], DocFormer [1], and StrucTexT [10], reading-order modeling has been integrated into multimodal representation learning. These methods are widely applied to OCR post-processing, document information extraction, and question answering. Recent advances further explore unified modeling of text, layout, and vision [4, 6–8, 15], improving performance on complex layouts. OCR systems such as TrOCR [9] and document parsing models such as Nougat [2] also benefit from improved ordering strategies.

However, existing studies mostly target relatively regular layouts. They pay limited attention to real handwritten documents, especially in educational settings. Handwritten documents often feature blurred block boundaries, local insertions, and cross-region writing. Simple geometric assumptions fail under such conditions. Moreover, current benchmarks operate at block- or page-level granularity, limiting insights into errors unique to complex-layout OCR. To address this gap, we focus on fine-grained, sequence-level reading-order analysis. Our approach distinguishes structural errors such as text interleaving, cross-question misalignment, and local reversals. This allows more precise evaluation of reading-order recovery for handwritten mathematical OCR on complex layouts.

2.2 Mathematical Document Information Extraction Benchmarks

Existing research on mathematical document understanding falls into three categories. The first category targets document parsing and math-content extraction, including benchmarks and models such as OmniDocBench [14], MathDoc [21], and layout-aware document understanding frameworks [3]. These benchmarks evaluate models on structure, text, and formula extraction. They advance assessment from character recognition toward structured understanding but mostly focus on printed, clean, and regular layouts. The second category addresses real-world visual math inputs and document understanding tasks, including datasets such as DocVQA [12]. These works highlight challenges from capture noise, occlusion, blur, and complex educational scenarios, emphasizing real-world robustness. The third category evaluates multimodal models on math images, including MathVista, MathVerse, MATH-Vision, and MV-MATH [11, 16, 17, 20]. These studies focus on problem-solving performance rather than faithful reconstruction of original document content or structure.

Despite these advances, systematic evaluation for handwritten mathematical documents in realistic educational settings remains limited. Unlike printed or ideal inputs, this scenario requires accurate character, formula recognition and correct reading-order recovery, while controlling hallucinated content under incomplete evidence. Existing benchmarks often merge content, structure, and order errors into a single metric, obscuring fine-grained failure patterns. We address this gap by systematically evaluating models on real handwritten high-school math answers under complex layout and formula-mixed conditions.

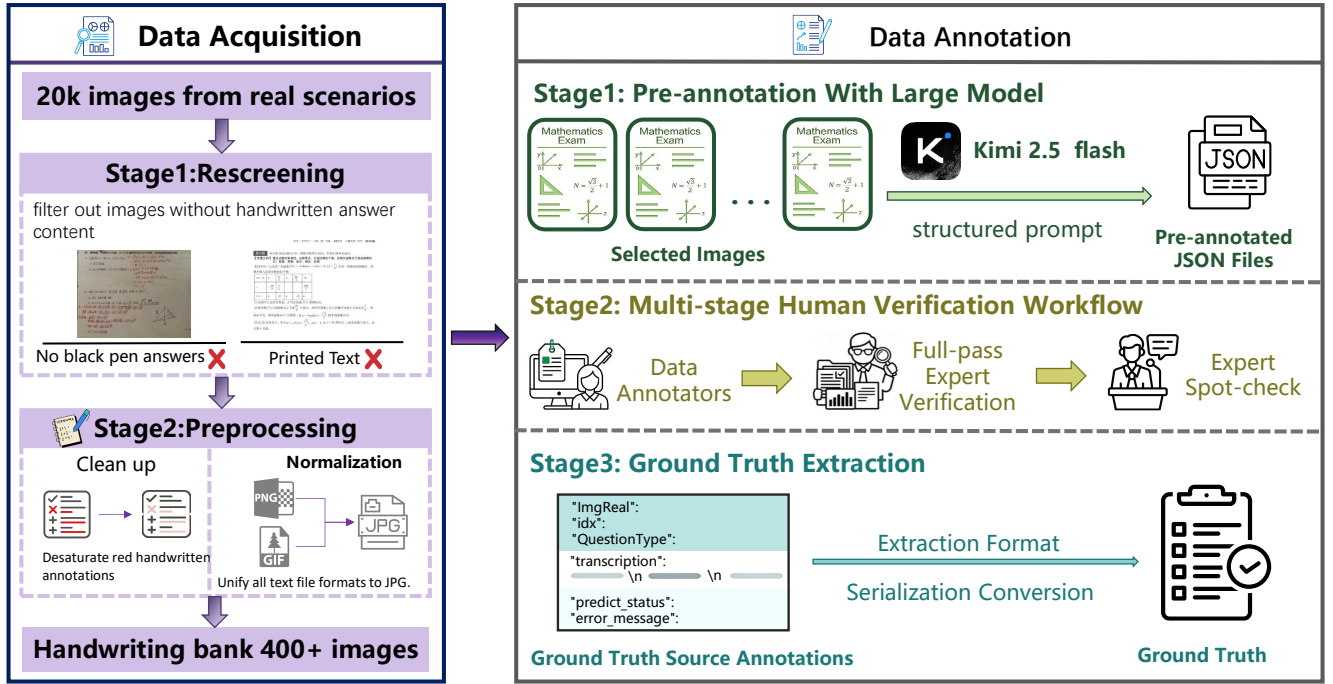


Figure 1: Dataset construction pipeline. A pipeline showing raw exam images, filtering steps, and final curated samples.

3 Dataset Construction

3.1 Data Acquisition

The dataset is collected from real Chinese high school math exams and contains rich handwritten characteristics, including diverse writing styles, cursive strokes, and irregular character shapes, as well as complex layouts such as multi-column and mixed text-image formats. It also includes real-world noise such as red annotations, blurred shadows, rotation, and partial cropping. These properties introduce joint challenges in visual recognition, structural parsing, and mathematical understanding, making the dataset a high-difficulty benchmark for handwritten answer OCR systems.

During dataset construction, we randomly sample 1,000 sheets from 60,000 raw Chinese high school math exam papers provided by Gaotu. The resulting dataset faithfully reflects real-world distributions, including natural noise under varying lighting conditions and diverse handwritten characteristics.

After screening and preprocessing, we obtain 446 valid samples (see Fig. 1). The process consists of two steps:

- **Content screening (Rescreening).** Printed-only sheets without handwritten responses are removed. Images containing only red-pen marks are also discarded to ensure the presence of valid handwritten content.
- **Preprocessing (Reprocessing).** Samples are standardized to reduce noise, including: (i) **overlaid mark attenuation**, where red-pen grading marks are attenuated to reduce interference; and (ii) **format normalization**, where all images are converted to JPG format to avoid OCR variations caused by inconsistent formats.

3.2 Data Annotation

To construct high-quality, structure-aware GT efficiently, we design a three-stage pipeline. The pipeline integrates pre-annotation, expert human annotation, and deterministic extraction.

- **Pre-annotation with MLLM.** The annotation pipeline starts with a multimodal large language model. We use Kimi 2.5 Flash to generate an initial transcription for each sample. A structured prompt constrains the output format. This step reduces manual annotation cost and improves recall. The output serves as a high-recall initializer for subsequent manual correction. The output is not treated as GT.
- **Expert human annotation.** Four trained annotators perform character-level manual correction under a strict protocol. Then, one expert performs full-pass verification over all samples to ensure consistency. A second expert conducts spot-checking on a subset of samples to identify systematic errors. The output of this process forms the **GT source annotations**.
- **Ground-truth extraction pipeline.** To enable direct evaluation, we apply a deterministic pipeline to convert unstructured annotations into structured GT. First, transcriptions are split by line breaks and assigned to sub-questions via temporary `[[qNNN]]` tags, preserving the original reading order. Second, expressions enclosed by `[[MATH: ...]]` are extracted into sample-specific lists for fine-grained evaluation. Ultimately, the final GT comprises three levels of representation: line-level structured transcriptions, extracted formula lists, and the unmodified final-answer string.

3.3 Dataset Properties

446 images produce 6,129 text sequences after serialization. Among them, 5,502 sequences are formulas (89.8%), while 627 sequences are non-formula text (10.2%).

The dataset comprises 446 questions, with an average of 415.4 characters per question and a maximum length of 1,765. The large length variation reflects dense interleaving of text and mathematical expressions.

This distribution presents high structural complexity and requires strong recognition and parsing capability.

4 Evaluation Methodology

4.1 Task Definition

We define the task as structured parsing and multi-dimensional evaluation of handwritten response content on mathematical exam papers. Given a handwritten input image I , the model must recognize text and formulas and produce a structured prediction record p aligned with GT. The task includes four subtasks: text extraction, formula recognition, reading order judgment, and intelligent refusal. These subtasks enable quantitative evaluation across multiple dimensions.

(1) **Text extraction.** The model performs line-level recognition of handwritten text. The predicted transcription sequence P must match the ground truth sequence G . The prediction must preserve semantic information and content integrity.

(2) **Formula recognition.** The model extracts mathematical expressions from the transcription in structured form. The predicted formula set F_{pred} must align with the ground truth set F_{gt} . The alignment requires consistency in symbol layout and operator logic.

(3) **Reading order judgment.** The predicted text and formula sequence must follow natural human reading order. The sequence must satisfy block continuity and correct sub-question order. The sequence must also preserve intra-question order. The evaluation supports quantitative scoring of order errors.

(4) **Intelligent refusal.** The model performs extraction only for recognizable samples. We define a ground truth-based function $\Phi(I) \in \{0, 1\}$ to judge recognizability. When $\Phi(I) = 0$, the sample is treated as a refusal case. This case includes three scenarios:

- **Key region occlusion:** Core text or formula regions are fully covered by corrections, stains, or artifacts.
- **Impaired physical integrity:** The image is truncated or damaged, which causes content loss or broken semantics.
- **Severe noise interference:** The image contains heavy blur, rotation, skew, or excessive annotation marks, which prevents effective parsing.

The model outputs a structured prediction record

$$p = \{\text{filename, idx, QuestionType, transcription, formula_list, final_answer, } y_{\text{pred}}\}$$

The label y_{pred} satisfies:

$$y_{\text{pred}} = \begin{cases} 1, & \text{if } \Phi(I) = 1, \text{ output complete structured content} \\ 0, & \text{if } \Phi(I) = 0, \text{ output only refusal label} \end{cases}$$

4.2 Evaluation Pipeline Construction

To ensure fair and comprehensive evaluation across different models, we design an end-to-end evaluation pipeline. The pipeline starts from prediction generation, then performs processing and alignment, and finally computes metrics (see Fig. 2).

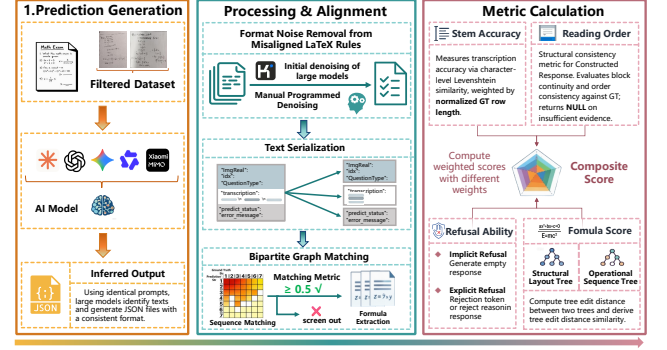


Figure 2: Overview of the evaluation pipeline.

4.2.1 **Extraction.** This stage corresponds to *Prediction Generation* and *Processing* in Fig. 2. For each image, we generate one raw prediction with a unified structured prompt and output schema. We then apply Kimi 2.5 for format normalization and noise attenuation. This step removes format-level noise and preserves semantic content. The processed output is

$$p = \psi(r). \quad (1)$$

The cleanup stage does not perform additional recognition. It only removes redundant wrappers and irregular formatting. We then obtain a unified structured prediction record:

$$p = \{\text{filename, idx, QuestionType, transcription, formula_list, final_answer}\} \quad (2)$$

For structured responses, we split the cleaned transcription by sub-question tags and line breaks. This step produces an ordered sequence set

$$T = \{t_1, t_2, \dots, t_n\}. \quad (3)$$

Each segment contains a sequence index, a sub-question identifier, and text content. For plain responses, transcription remains a single string. This stage maps noisy outputs to deterministic structured predictions.

4.2.2 **Matching Algorithm.** This stage corresponds to *Alignment* in Fig. 2. We first pair samples by keys. We denote the ground truth sequence as

$$G = \{g_1, g_2, \dots, g_M\} \quad (4)$$

and the predicted sequence as

$$P = \{p_1, p_2, \dots, p_N\}. \quad (5)$$

We normalize each line with a function $\phi(\cdot)$ to remove non-semantic variation. We then compute normalized Levenshtein similarity

$$s_{ij} = 1 - \frac{\text{Lev}(\phi(g_i), \phi(p_j))}{\max(|\phi(g_i)|, |\phi(p_j)|)}. \quad (6)$$

We construct a similarity matrix $S \in \mathbb{R}^{M \times N}$ and solve a one-to-one bipartite matching problem

$$A^* = \arg \max_A \sum_i \sum_j a_{ij} s_{ij}, \quad (7)$$

subject to a matching threshold $\tau = 0.5$. Matches below the threshold are discarded. Unmatched predictions are treated as noise. Unmatched ground truth lines are treated as missing content.

For matched text segments, we extract formulas from transcription as a set

$$F = \{f_1, f_2, \dots, f_m\}. \quad (8)$$

We align formulas in a slot-wise manner within each sub-question. Missing slots indicate missing formulas, and extra slots indicate redundant formulas. This design provides deterministic alignment for formula evaluation.

We also define refusal alignment at the sample level. Let y_{gt} and y_{pred} denote binary labels. If either label indicates an unreadable sample, we skip text and formula evaluation and only assess refusal behavior.

4.2.3 Metric Calculation. This stage corresponds to *Evaluation* in Fig. 2. We compute both sample-level and dataset-level metrics.

Text extraction. We compute Stem Accuracy. For each ground truth line g_i , we assign a score c_i . Matched lines receive similarity scores, and unmatched lines receive zero. We weight each line by its length:

$$w_i = |\phi(g_i)|. \quad (9)$$

The sample-level score is

$$\text{StemAcc} = \frac{\sum_{i=1}^M w_i c_i}{\sum_{i=1}^M w_i}. \quad (10)$$

Formula recognition. We evaluate formulas with two tree representations. The Symbol Layout Tree captures spatial structure. The Operator Tree captures semantic structure. For each aligned slot, we compute

$$\text{TEDS}(T_1, T_2) = 1 - \frac{\text{TED}(T_1, T_2)}{|T_1| + |T_2|}. \quad (11)$$

We combine layout and operator scores:

$$\text{FS}_k = 0.4 \cdot \text{SLT}_k + 0.6 \cdot \text{OPT}_k. \quad (12)$$

The sample-level score is the average over all slots.

Reading order. We evaluate reading order using matched coverage and sequence consistency. We compute a score based on block continuity, sub-question order, and intra-question order. The final score penalizes fragmented sequences and local disorder.

Intelligent refusal. We compare y_{gt} and y_{pred} and compute Precision, Recall, and F1. We also report hallucination rate as the ratio of missed valid samples.

At the dataset level, we aggregate metrics across all valid samples. Text scores use length-weighted averaging. Formula scores use slot-weighted averaging. Reading order scores are averaged over valid samples. Refusal metrics are computed from global counts.

Finally, we compute a composite score:

$$\begin{aligned} \text{Composite} = & 0.40 \cdot \text{OverallStemAcc} \\ & + 0.30 \cdot \text{OverallFormulaScore} \\ & + 0.20 \cdot \text{OverallROS} \\ & + 0.10 \cdot \text{OverallRefusalF1}. \end{aligned} \quad (13)$$

If a module is not applicable, we renormalize the weights over valid components.

5 Experiments

5.1 Main Result Analysis

5.1.1 Overall Conclusion. In terms of the Composite metric, model performance exhibits clear stratification. This demonstrates that the benchmark effectively differentiates the OCR capabilities of various models, validating the rationale behind its design.

5.1.2 Multi-dimensional Analysis. Text Extraction: The StemAcc metric achieves the best overall performance, indicating that mainstream multimodal models have developed relatively mature pure-text recognition capabilities. Notably, this metric primarily measures the extraction accuracy of the textual content itself and is relatively decoupled from reading-order modeling capabilities. Consequently, it serves as a purer reflection of the models' fundamental OCR proficiency in character identification, word restoration, and text transcription [18].

Reading Order: The reading-order metrics exhibit a characteristic pattern of being "locally easy but globally difficult". The average SQA and IQA scores reach as high as 0.9757 and 0.9787, respectively, indicating that local sequence-swapping and incorrect-insertion errors are minimal and have a limited impact on overall performance. In contrast, the average BCS is significantly lower at 0.7149, with ROS and ROS-Cov. averaging only 0.7356. This demonstrates that the primary bottleneck for models in real-world handwritten math scenarios is interleaving errors; namely, models are highly prone to erroneously integrating what should be parallel or subordinate content into the main solution chain. Even models with strong local sorting capabilities struggle to compensate for this issue. Therefore, suppressing interleaving errors is the key to improving the quality of reading-order recovery.

Intelligent Refusal: Refusal behavior (Refusal-F1) yields the lowest overall scores, reflecting severe hallucination issues in most models during OCR tasks. When confronted with blurred handwriting or complex layouts, models typically fail to make conservative refusal decisions, instead leaning toward generating superficially complete recognition outputs. Consequently, their errors manifest not only as character-level misrecognitions but also as structural-level hallucinations—specifically, the erroneous construction of global reading orders and content organizational relationships. It is worth noting that high composite performance does not equate to superior refusal control capabilities, indicating that the ability to control refusals has not yet become a stable component of current models' document understanding proficiency.

5.2 Semantic Sequence Rectification

The main results show a clear drop in reading order accuracy on complex handwritten layouts. The degradation is severe in samples

Table 1: Main benchmark results. Metrics follow a hierarchical structure: StemAcc, Formula, ROS, and Refusal-F1 constitute the four first-level components of the final Composite score. Within the Formula module, SLT-TEDS and OPT-TEDS are intermediate metrics aggregated into Formula; within ROS, BCS, SQA, and IQA are aggregated into ROS, with ROS-Cov. indicating the coverage of evaluable samples. Best values in each column are shown in bold.

Model ↓ / Metric →	Text Extraction	Formula Recognition			Reading Order				Refusal Behavior		Overall
	StemAcc	SLT-TEDS	OPT-TEDS	Formula	BCS	SQA	IQA	ROS	ROS-Cov.	Refusal-F1	Composite
kimi-k2.5	0.9120	0.7201	0.7197	0.7198	0.8549	0.9818	0.9917	0.8194	0.9428	0.0000	0.8520
qwen3.5-plus	0.8787	0.4657	0.4654	0.4655	0.8086	0.9833	0.9880	0.7899	0.9100	0.0426	0.8258
gemini-2.5-flash	0.8794	0.5555	0.5548	0.5551	0.7783	0.9809	0.9868	0.7386	0.8928	0.1250	0.8149
doubao-1.5-vision-pro-32K	0.8469	0.3241	0.3238	0.3239	0.6898	0.9909	0.9922	0.7612	0.7706	0.0444	0.7963
doubao-seed-1.6-vision	0.8376	0.3580	0.3584	0.3582	0.6067	0.9856	0.9919	0.7684	0.6802	0.3279	0.7958
qwen3-vl-plus	0.8462	0.2702	0.2698	0.2700	0.7570	0.9818	0.9902	0.7318	0.8234	0.0000	0.7871
claude-sonnet-4-5-20250929	0.8348	0.4984	0.4970	0.4976	0.7401	0.9632	0.9759	0.7177	0.7895	0.0870	0.7761
qwen3.5-flash	0.8372	0.2016	0.2015	0.2015	0.7637	0.9776	0.9913	0.7486	0.8333	0.0000	0.7653
gpt-4o	0.7859	0.2036	0.2032	0.2033	0.6528	0.9635	0.9612	0.6591	0.6119	0.0000	0.7206
grok-4-0709	0.6710	0.0833	0.0828	0.0830	0.4967	0.9483	0.9176	0.6215	0.1016	0.2010	0.6437
Average	0.8330	0.3681	0.3676	0.3678	0.7149	0.9757	0.9787	0.7356	0.7356	0.0828	0.7778

with interleaving, jumps, and cross-region transitions. The BCS metric drops the most in these cases. This phenomenon has two possible causes. First, OCR errors may corrupt the text. Second, the model may fail to recover the correct order even with correct text.

This section isolates these factors with a controlled experiment. We fix the OCR output and test whether an MLLM can restore the correct order using semantics alone. The goal is to evaluate semantic understanding and contextual modeling under identical text input. **We hypothesize the following.** For simple linear layouts, spatial order aligns with the logical solution flow. Both spatial sorting and MLLM reordering should achieve similar BCS. For complex non-linear layouts, spatial cues become unreliable. These layouts include multi-column writing, arrow-based jumps, inline insertions, and mixed draft regions. An MLLM with strong semantic reasoning should exploit contextual continuity and step dependencies to recover a better order.

5.2.1 Experimental Setup. We collect 40 samples from real high school handwritten math exams and homework. We annotate the ground truth reading order for each sample. We control image quality to remove confounding factors. All samples are clear, legible, and free of occlusion.

We split the data into two subsets based on layout complexity:

- **Simple Linear Layout** ($N = 20$). Each sample uses a single column. The solution follows a top-to-bottom and left-to-right flow. No cross-region insertions or arrow jumps appear.
- **Complex Non-linear Layout** ($N = 20$). Samples include multi-column writing, arrow-guided jumps, inline insertions, and mixed draft regions. The physical writing order deviates from the logical solution order.

We use PaddleOCR-VL to extract text units for all samples. This step ensures identical OCR input for all methods. We then compare two ordering strategies:

- (1) **Spatial OCR Baseline.** We use the reading order produced by the OCR system. This method relies on spatial proximity and layout structure.
- (2) **MLLM-based Reordering.** We input the same OCR text units into an MLLM. We instruct the model to reorder based on semantic progression, coherence, and step dependencies, rather than spatial position.

Both methods share the same OCR output. Therefore, performance differences reflect sequence reconstruction ability rather than recognition quality. Prior analysis shows that interleaving and jumps dominate errors in complex layouts. We thus use average BCS as the sole metric.

5.2.2 Result Analysis. Table 2 reports BCS under identical OCR input. On simple linear layouts, both methods achieve high scores. The gap is only 0.8 points. This result shows that spatial cues suffice when layout matches logical flow. The semantic advantage of the MLLM is limited in this case.

On complex non-linear layouts, the gap increases sharply. The spatial baseline drops to 0.628, while MLLM reordering reaches 0.849. This result shows that spatial heuristics fail under complex structures. The MLLM leverages semantic relations and solution logic to correct disrupted sequences.

The performance drop from simple to complex layouts is much larger for the spatial baseline. This pattern indicates higher sensitivity to layout complexity. In contrast, the MLLM shows stronger robustness.

Both methods use identical OCR outputs. Therefore, the improvement cannot come from better recognition. The gain reflects superior semantic understanding. These results confirm that, with fixed OCR quality, MLLMs can effectively rectify reading order errors in complex layouts.

Table 2: Quantitative comparison of multimodal models on the chart understanding benchmark. Best results within each subset are shown in bold.

Method	Subset	Avg. BCS $\uparrow$	Avg. SQA $\uparrow$	Avg. IQA $\uparrow$	Avg. ROS $\uparrow$	ROS Cov. $\uparrow$
Claude Sonnet 4.5	Simple Linear	0.9800	1.0000	0.9851	0.9989	0.6000
	Complex Non-linear	0.8561	0.9683	0.9238	0.9278	0.5238
	Overall	0.9166	0.9837	0.9537	0.9649	0.5610
Doubao 1.5 Vision Pro 32K	Simple Linear	0.9550	0.9500	0.9903	0.9622	0.6000
	Complex Non-linear	0.8392	0.9524	0.9205	0.8500	0.5238
	Overall	0.8957	0.9512	0.9545	0.9086	0.5610
Doubao Seed 1.6 Vision	Simple Linear	0.9800	1.0000	0.9906	0.9998	0.6000
	Complex Non-linear	0.9081	1.0000	0.9926	0.9981	0.4000
	Overall	0.9656	1.0000	0.9910	0.9996	0.5600
Gemini 2.5 Flash	Simple Linear	0.9789	1.0000	0.9730	0.9969	0.5789
	Complex Non-linear	0.8598	0.9474	0.9124	0.8828	0.5263
	Overall	0.9194	0.9737	0.9427	0.9426	0.5526
GPT-4o	Simple Linear	0.9800	0.9500	0.9459	0.9674	0.6000
	Complex Non-linear	0.8044	0.9048	0.8403	0.7837	0.5238
	Overall	0.8901	0.9268	0.8918	0.8796	0.5610
Kimi-K2.5	Simple Linear	0.9800	1.0000	0.9818	0.9984	0.6000
	Complex Non-linear	0.9426	1.0000	0.9592	0.9935	0.6250
	Overall	0.9693	1.0000	0.9753	0.9969	0.6071
MIMO-V2-Omni	Simple Linear	0.9800	1.0000	0.9873	0.9993	0.6000
	Complex Non-linear	0.8476	0.9683	0.9234	0.9432	0.5238
	Overall	0.9122	0.9837	0.9546	0.9725	0.5610
Qwen3-VL-Plus	Simple Linear	0.9800	0.9500	0.9768	0.9975	0.6000
	Complex Non-linear	0.8397	0.9683	0.9051	0.8892	0.5238
	Overall	0.9081	0.9593	0.9401	0.9457	0.5610
Qwen3.5-Flash	Simple Linear	0.9800	1.0000	0.9894	0.9996	0.6000
	Complex Non-linear	0.8546	0.9524	0.9266	0.8942	0.5238
	Overall	0.9158	0.9756	0.9572	0.9492	0.5610
Qwen3.5-Plus	Simple Linear	0.9653	0.9000	0.9722	0.9718	0.6000
	Complex Non-linear	0.8275	0.9524	0.9216	0.8447	0.5238
	Overall	0.8947	0.9268	0.9463	0.9110	0.5610
Spatial OCR Baseline	Simple Linear	0.9800	1.0000	0.9917	1.0000	0.6000
	Complex Non-linear	0.7251	0.9524	0.9267	0.5002	0.5238
	Overall	0.9158	0.9756	0.9584	0.9506	0.5610

5.3 Decoupled Inference for OCR Hallucination Mitigation

5.3.1 Methodology. To mitigate OCR hallucinations in MLLMs caused by the over-reliance on language priors under insufficient visual evidence, we propose a **decoupled inference** framework. This approach decomposes the OCR process into two distinct stages: **(1) Recognizability Assessment**, and **(2) Text Generation**. In this paradigm, the model first performs an explicit judgment on image quality; the OCR generation is triggered only if the image is deemed "recognizable." Otherwise, the system issues a rejection response. To evaluate its efficacy, we conducted a comparative experiment using 40 manually annotated "unrecognizable" images, benchmarking the decoupled pipeline against the standard **coupled inference** (where perception and generation occur in a single pass).

5.3.2 Results and Discussion. Experimental results demonstrate that **decoupled inference significantly reduces the hallucination rate** compared to the coupled baseline. Specifically, by isolating the assessment of visual clarity from the generative task, the model effectively suppresses the tendency to "hallucinate" plausible but unfounded text when visual cues are absent. This suggests

that explicit logical decoupling strengthens the model’s rejection capability and mitigates the interference of internal language priors. Our findings provide a simple yet robust strategy for improving the faithfulness of MLLM-based OCR systems in real-world scenarios involving degraded inputs.

6 Semantic-Guided Reading-Order Rectification Agent

Our diagnostic experiments on MathScript reveal clear structural parsing failures in complex handwritten layouts. When layouts shift from linear to nonlinear structures, models produce errors in region interleaving, cross-region jumps, and local sequence connections. These errors reduce reading-order metrics. SQA and IQA remain high, but BCS is much lower. This pattern shows that the main bottleneck lies in suppressing interleaving errors and restoring global logical chains.

To address this issue, we propose a semantic-guided Reading-Order Rectification Agent (RRA). The agent reorders predicted sequences without modifying OCR content. It improves structural consistency in complex layouts.

Table 3: Benchmark results. The table preserves the same metric hierarchy as Table 1. StemAcc, Formula, ROS, and Refusal-F1 are the four first-level components of the final Composite score. For each entry, if the updated value is larger than its previous counterpart, the improvement is appended in parentheses as $x(+\Delta)$. The four first-level component columns are boldfaced for all models, and the best value in each column is underlined.

Model ↓ / Metric →	Text Extraction	Formula Recognition			Reading Order				Refusal Behavior		Overall
	StemAcc	SLT-TEDS	OPT-TEDS	Formula	BCS	SQA	IQA	ROS	ROS-Cov.	Refusal-F1	Composite
kimi-k2.5	0.9120	<u>0.9241</u>	<u>0.9236</u>	0.9238	0.9867(+0.1318)	0.9818	0.9917	0.9852(+0.1658)	<u>0.9428</u>	0.0000	0.9323(+0.0331)
qwen3.5-plus	0.8787	0.8981	0.8976	0.8978	0.9759(+0.1673)	0.9833	0.9880	0.9762(+0.1863)	0.9100	0.0426	0.9065(+0.0373)
gemini-2.5-flash	0.8794	0.8841	0.8829	0.8834	0.9266(+0.1483)	0.9809	0.9868	0.9223(+0.1837)	0.8928	0.1250	0.8901(+0.0368)
doubao-1.5-vision-pro-32K	0.8469	0.8664	0.8656	0.8660	0.8489(+0.1591)	<u>0.9909</u>	<u>0.9922</u>	0.9704(+0.2092)	0.7706	0.0444	0.8745(+0.0419)
doubao-seed-1.6-vision	0.8376	0.8302	0.8309	0.8306	0.7431(+0.1364)	0.9856	0.9919	0.9786(+0.2102)	0.6802	<u>0.3279</u>	0.8555(+0.0421)
qwen3-vl-plus	0.8462	0.8642	0.8632	0.8636	0.9737(+0.2167)	0.9818	0.9902	0.9721(+0.2403)	0.8234	0.0000	0.8763(+0.0480)
claude-sonnet-4-5-20250929	0.8348	0.8423	0.8399	0.8409	0.9476(+0.2075)	0.9632	0.9759	0.9504(+0.2327)	0.7895	0.0870	0.8578(+0.0465)
qwen3.5-flash	0.8372	0.8025	0.8019	0.8021	0.9729(+0.2092)	0.9776	0.9913	0.9740(+0.2254)	0.8333	0.0000	0.8521(+0.0451)
gpt-4o	0.7859	0.7848	0.7834	0.7839	0.8981(+0.2453)	0.9635	0.9612	0.9315(+0.2724)	0.6119	0.0000	0.8056(+0.0545)
grok-4-0709	0.6710	0.6909	0.6869	0.6885	0.6510(+0.1543)	0.9483	0.9176	0.8835(+0.2620)	0.1016	0.2010	0.6912(+0.0524)
Average	0.8330	0.8388	0.8376	0.8381	0.8925	0.9757	0.9787	0.9544	0.7356	0.0828	0.8215

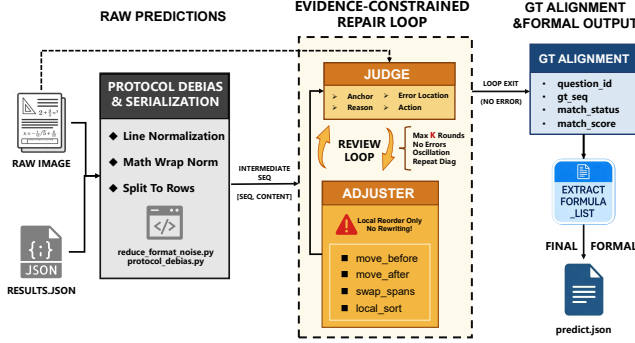


Figure 3: Overview of the proposed RRA framework.

We model reading-order repair as a constrained sequence re-ordering problem. Let the model produce a sequence

$$X^{(0)} = \{x_1, x_2, \dots, x_n\},$$

where each element x_i contains a predicted position seq and text $content$. The goal is not to regenerate text or align with ground truth early. Instead, we adjust the relative order of elements and obtain a new sequence X^* . The content remains unchanged. This design decouples order repair from GT alignment. It provides a stable intermediate sequence for downstream processing.

The RRA is inserted between sequence serialization and GT alignment. The model output first undergoes noise cleaning and normalization. Then the system splits it into structured transcription lines. The RRA iteratively reviews and adjusts this sequence. After convergence, the system performs GT alignment and extracts *formula_list* for evaluation. This module does not replace the evaluation logic. It is fully plug-and-play.

We design a two-stage closed-loop agent: Judge–Edit–Review.

Figure 3 shows the pipeline. The first stage is the Judge. It takes the image I , the current sequence $X^{(t)}$, and the history H . The Judge

does not generate a new sequence. Instead, it detects anomalies and outputs a structured diagnosis $D^{(t)}$. The diagnosis includes error flags, suspicious segments, action suggestions, anchor positions, and reasoning. This design separates detection from modification and avoids uncontrolled drift.

The second stage is the Editor. It takes $X^{(t)}$, $D^{(t)}$, and I . It performs minimal edits within a constrained action space. Allowed operations include *move_before*, *move_after*, *swap_spans*, and *local_sort*. The Editor only reorders segments and never modifies text content. It produces a candidate sequence $Y^{(t+1)}$. This constraint ensures interpretability and traceability.

After editing, the Judge reviews the candidate sequence. If no anomalies remain, the loop stops. Otherwise, the system continues to the next iteration. We use discrete termination conditions. The process stops when the iteration limit K is reached, when no error is detected, when diagnoses repeat, or when oscillation occurs. This strategy prevents unstable loops and ensures efficiency.

We evaluate the proposed RRA on MathScript. As shown in Table 3, both BCS and ROS improve significantly compared to the baseline. This result shows that RRA effectively uses semantic information to repair reading order at the sequence level.

7 Conclusion

In this paper, we introduce **MathScript**, a fine-grained benchmark for evaluating MLLMs on handwritten mathematical content recognition in realistic educational scenarios. Through comprehensive evaluation and diagnostic analysis, we identify two major failure modes of current models. First, models show structural parsing errors in complex non-linear layouts. Second, models exhibit unreliable behavior under insufficient visual evidence, including hallucinations and imbalance in refusal decisions. Guided by these findings, we further explore benchmark-driven improvement strategies and verify their effectiveness on both dimensions. We hope **MathScript** can serve as a useful testbed for developing more faithful, reliable, and structure-aware multimodal document understanding systems.

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